



VJEDAI

V-JEPA Encoder for Depth Anything Inference

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Abstract

We introduce VJEDAI (V-JEPA Encoder for Depth Anything Inference), which couples a frozen, video-native V-JEPA 2.1 encoder to a trainable Depth Anything V2 decoder with a per-pixel uncertainty head. This reuses existing perception backbones while producing confidence estimates for safety, explainability, and distillation.

1. Introduction

Monocular depth estimation has shifted from highly supervised, domain-specific CNN regression toward large pre-trained vision transformers with stronger geometric representations (Dosovitskiy et al., 2021). In this work, we introduce VJEDAI, a hybrid monocular depth model that replaces the Depth Anything V2 image encoder with a frozen V-JEPA 2.1 backbone while retaining the Depth Anything DPT decoder for dense depth prediction.

A key motivation for this design is that V-JEPA is video-native: its input representation includes an explicit temporal dimension, allowing the same architecture to naturally extend from single-frame depth estimation to video-based perception. This differs from the Depth Anything family, where video-consistent depth requires further architectural extensions on top of the image-based model (Chen et al., 2025).

The model also avoids redundant visual encoding. A specialized Depth Anything V2-style system would still rely on a DINOv2 backbone, which is usually fine-tuned and maintained separately for depth. In contrast, VJEDAI freezes the V-JEPA encoder and trains only the Depth Anything

decoder and uncertainty head. This is especially useful for robotics platforms that already run V-JEPA as part of their perception stack, since depth estimation can reuse the same backbone instead of adding another large encoder.

Beyond predicting depth, VJEDAI outputs per-pixel uncertainty values through a Gaussian uncertainty head. These uncertainty estimates can support safer deployment by highlighting unreliable predictions, improve explainability by exposing where the model is less confident, and provide useful supervision signals for distilling the model into a smaller student network. During training, the uncertainty head also allows the model to account for noisy or ambiguous pixels, which can, as we have observed, improve generalization compared with optimizing depth error alone.

2. Models and Methods

2.1. Pretrained Vision Backbones for Depth Estimation

One example of the shift from CNNs to transformer architecture for monocular depth estimation is Depth Anything V2 (DA2). It uses a dense prediction transformer (DPT) decoder on top of DINOv2 encoders, trains a reliable teacher model on accurate synthetic depth, creates pseudo depth for large-scale unlabeled real images, and finally trains student models for robust cross-domain generalization (Yang et al., 2024).

Architecturally, DA2 uses a DPT-style decoder: intermediate DINOv2 token sequences are reshaped into spatial feature maps, projected into a multi-scale feature pyramid, fused from coarse to fine with RefineNet-style blocks, and upsampled to produce the final depth map (DepthAnything, 2026). Depth Anything 3 (DA3) keeps a transformer backbone but extends the prediction head to a Dual-DPT design, where shared features are split into separate branches for depth and ray-map prediction, optionally using camera information for additional geometric context (Lin et al., 2025).

While the Depth Anything family relies on DINOv2, an image-based self-supervised encoder, V-JEPA 2.1 offers an alternative backbone learned from video. It is a self-supervised video representation model designed to learn

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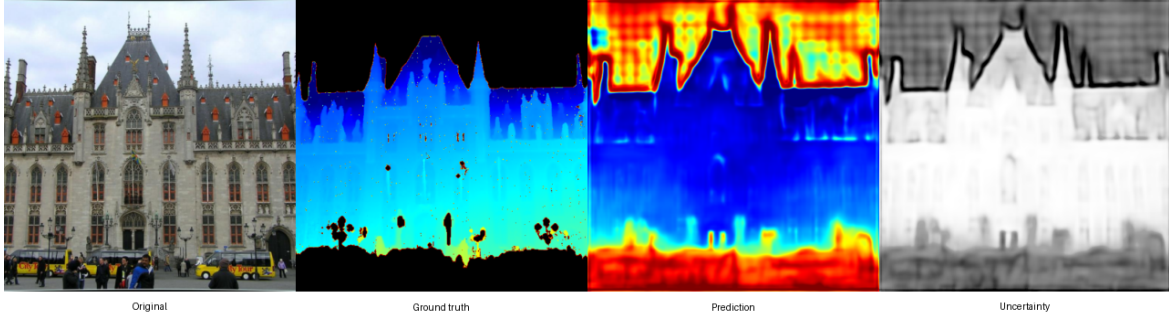


Figure 1. Example RGB input and predicted output. For uncertainty, the darker, the less certain.

dense visual features for both images and videos. Rather than reconstructing raw pixels, it learns through latent feature prediction, encouraging spatially structured and temporally consistent representations. This makes it particularly relevant for monocular depth estimation, where the model must recover geometric structure from visual cues. The V-JEPA 2.1 authors further show that a simple linear probe trained on top of the frozen encoder performs well on NYUv2 depth estimation, indicating that depth-relevant information is already present in the learned representation (Mur-Labadia et al., 2026). This motivates our use of V-JEPA 2.1 as the encoder backbone in VJEDAI.

2.2. VJEDAI Architecture

VJEDAI combines a frozen V-JEPA 2.1 visual encoder with the DPT decoder used in Depth Anything V2. Since the selected V-JEPA and DINOv2 variants have matching feature dimensions, the DPT decoder can consume reshaped V-JEPA token features directly. This removes the original DINOv2 encoder from Depth Anything while preserving the decoder structure described above.

Given an input image $\mathbf{I}$, the V-JEPA encoder produces intermediate token features from four transformer layers. In the large variant, these layers are [5, 11, 17, 23]. Each token sequence is reshaped into a spatial feature map and passed through the projection and resizing layers of the Depth Anything DPT head. The resulting multi-scale features are fused from coarse to fine, producing a dense feature map for prediction.

The final model outputs both a dense depth map and a per-pixel log-variance map:

$$f(\mathbf{I}) = \{\hat{\mathbf{D}}, \log \sigma^2\},$$

where $\hat{\mathbf{D}}$ is the predicted depth and $\log \sigma^2$ represents the predicted uncertainty at each pixel. The uncertainty head is attached to the shared decoder feature map before the final depth projection and consists of a small convolutional prediction branch aligned with the depth output.

2.3. Dataset and Preprocessing

For all experiments, we use the training data provided in the Kaggle competition [ETHZ CIL Monocular Depth Estimation 2026](#), which consists of around 23K RGB images, each paired with a depth map saved as a numpy array. We split the data into a training split (90%) and a validation split (10%), using a fixed random seed for reproducibility.

Input RGB images are converted to floating-point tensors and normalized with ImageNet-style channel statistics, matching the preprocessing used by Depth Anything and V-JEPA-based vision backbones. For VJEDAI, each image is resized to 384×384 , normalized using the values in Table 1, and reshaped into a video-style tensor by adding a singleton temporal dimension. Thus, an image batch is transformed from shape $[B, 3, H, W]$ to $[B, 3, 1, 384, 384]$ before being passed to the V-JEPA encoder. The predicted depth map is then upsampled back to the original image resolution.

Statistic	Red	Green	Blue
Mean	0.485	0.456	0.406
Standard deviation	0.229	0.224	0.225

Table 1. Mean and standard deviation used to normalize the input images.

2.4. Training Objective

VJEDAI is trained with a scale-invariant objective in log-depth space. Let $\hat{D}_{i,p}$ be the predicted depth at pixel p in image i , and let $D_{i,p}$ be the corresponding ground-truth depth. For the valid pixel set Ω_i , we define the log-depth residual as

$$d_{i,p} = \log(\max(\hat{D}_{i,p}, \epsilon)) - \log(D_{i,p}),$$

where $\epsilon = 10^{-6}$ prevents numerical instability. Since monocular depth is ambiguous up to scale, we remove the mean residual independently for each image:

$$r_{i,p} = d_{i,p} - \frac{1}{|\Omega_i|} \sum_{q \in \Omega_i} d_{i,q}.$$

The deterministic depth model is trained using scale-invariant RMSE:

$$\mathcal{L}_{\text{SI-RMSE}} = \frac{1}{N} \sum_i \sqrt{\frac{1}{|\Omega_i|} \sum_{p \in \Omega_i} r_{i,p}^2}.$$

This objective directly matches the validation metric used for model selection and optimizes the depth prediction without using the uncertainty output.

We also train a separate uncertainty-aware model with a heteroscedastic Gaussian negative log-likelihood using the predicted log-variance $s_{i,p} = \log \sigma_{i,p}^2$. The loss is

$$\mathcal{L}_{\text{NLL}} = \frac{1}{\sum_i |\Omega_i|} \sum_i \sum_{p \in \Omega_i} \frac{1}{2} (\tilde{s}_{i,p} + r_{i,p}^2 \exp(-\tilde{s}_{i,p})),$$

where $\tilde{s}_{i,p}$ is the clipped log-variance. In our implementation, $s_{i,p}$ is clipped to the range $[-7, 3]$ to prevent the model from reducing the loss by predicting excessively large uncertainty.

3. Results

3.1. Evaluation Metrics

We evaluate depth accuracy using AbsRel, SI-RMSE, and δ_1 . Absolute relative error is defined as

$$\text{AbsRel} = \frac{1}{|\Omega|} \sum_{p \in \Omega} \frac{|\hat{D}_p - D_p|}{D_p}.$$

δ_1 measures the fraction of valid pixels whose prediction is within a threshold of the ground truth:

$$\delta_1 = \frac{1}{|\Omega|} \left| \left\{ p \in \Omega : \max \left(\frac{\hat{D}_p}{D_p}, \frac{D_p}{\hat{D}_p} \right) < 1.25 \right\} \right|.$$

The uncertainty metrics in Table 5 are computed from the zero-mean log-depth residual $r_{i,p}$ and predicted standard deviation

$$\sigma_{i,p} = \exp(\tilde{s}_{i,p}/2), \quad z_{i,p} = \frac{r_{i,p}}{\sigma_{i,p}}.$$

Norm-std is the empirical standard deviation of $z_{i,p}$, which should be close to one for calibrated uncertainty. RMSE / RMS-sigma compares observed error scale to predicted uncertainty scale. ECE is the mean absolute gap between nominal Gaussian coverage levels $\{50, 68, 90, 95\}\%$ and empirical coverage. Corr(abs err, sigma) is the Pearson correlation between $|r_{i,p}|$ and $\sigma_{i,p}$.

3.2. Baselines and Decoder Selection

We first train a per-patch linear probe on the final V-JEPA outputs to verify that the frozen representation contains

depth information. The probe is trained for 3 epochs and achieves an SI-RMSE of 0.66 on the validation set, confirming that V-JEPA provides a useful depth signal.

For the decoder experiments, we freeze the encoder and train the DPT head under two settings: starting from pretrained weights, and starting from randomly re-initialized weights. All runs use AdamW with learning rate $5 \cdot 10^{-5}$, batch size 8, and 10 epochs on the 90% training split.

Model	AbsRel	si-RMSE	δ_1
DA2 pretrained	13.557	0.530	0.047
DA2 finetuned	2.202	0.552	0.286
DA2 trained	2.353	0.550	0.289
DA3 pretrained	4.515	0.523	0.125
DA3 finetuned	1.966	0.512	0.724
DA3 trained	2.377	0.605	0.187

Table 2. Comparison of pretrained and fine-tuned Depth Anything decoders on the validation split. “Pretrained” denotes zero-shot evaluation, “finetuned” denotes training only the DPT head initialized from pretrained weights, and “trained” denotes training only a randomly re-initialized DPT head while keeping the encoder frozen. Lower AbsRel and si-RMSE are better, while higher δ_1 is better.

Table 2 shows that both DA2 and DA3 benefit from decoder adaptation. The pretrained models have poor AbsRel and δ_1 , indicating that their absolute depth scale does not transfer well without fine-tuning. DA3 finetuning gives the best standalone validation result, with AbsRel 1.966, si-RMSE 0.512, and $\delta_1 = 0.724$.

However, DA2 is more stable under decoder re-initialization: its finetuned and randomly trained heads perform similarly, while DA3 degrades sharply when its head is re-initialized. Since our final model modifies the decoder and adds an uncertainty head, this robustness is more important than the best isolated score. DA2 is also simpler to integrate, slightly smaller, and does not rely on DA3’s ray-map supervision, which is unavailable in the competition dataset. We therefore use the DA2 decoder initialized from pretrained weights for the final model.

3.3. Training Results

Training objective	Train	Val	Test
SI-RMSE	0.4441	0.4572	0.570
SI-Gaussian NLL	0.4661	0.4672	0.500

Table 3. Final fine-tuning results after 40 epochs. All values are SI-RMSE.

The 10-epoch comparison shows that training JEDAI from scratch is less reliable than adapting a pretrained DPT head, so we keep the pretrained initialization for the final model. In the final 40-epoch runs, SI-Gaussian NLL achieves the

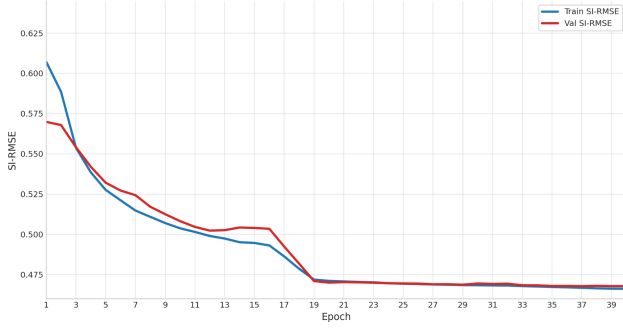


Figure 2. Training logs for the final VJEDAI run.

Initialization	Objective	Train	Val
Pretrained	SI-RMSE	0.4910	0.4940
Scratch	SI-RMSE	0.5568	0.3520
Pretrained	SI-Gaussian NLL	0.5038	0.5091
Scratch	SI-Gaussian NLL	0.5582	0.3499

Table 4. Short 10-epoch comparison of DPT head initialization and training objective. All values are SI-RMSE.

best test SI-RMSE despite a slightly worse validation SI-RMSE than direct SI-RMSE training. We attribute this to noise in the dataset: modeling heteroscedastic uncertainty makes the objective less dominated by ambiguous or noisy pixels, improving generalization on the test set.

3.4. Calibration results

Metric	Value
Norm-std	1.0365
RMSE / RMS-sigma	0.9851
ECE	0.1381
Corr(abs err, sigma)	0.3630
Coverage 50%	0.7647
Coverage 68%	0.8840
Coverage 90%	0.9606
Coverage 95%	0.9729

Table 5. Uncertainty calibration and coverage metrics for the final model at epoch 20.

Table 5 shows that the uncertainty estimates are reasonably well calibrated after Gaussian NLL training. The normalized residual standard deviation is close to one, and RMSE / RMS-sigma is 0.9851, indicating that the predicted uncertainty scale matches the observed error scale. The positive correlation between absolute error and predicted uncertainty shows that the model assigns higher uncertainty to harder pixels. Coverage is above the nominal levels at all reported thresholds, so the uncertainty estimates are slightly conservative, which is also reflected in the ECE of 0.1381. In our

experiments, training beyond 20 epochs gave only marginal gains in training loss but degraded calibration, so we select the deployable checkpoint at epoch 20, by balancing test SI-RMSE with uncertainty quality. Further calibration could be obtained through temperature rescaling of the predicted variances.

4. Discussion

VJEDAI shows that a frozen, video-native V-JEPA encoder can be adapted for dense monocular depth prediction using a Depth Anything V2 decoder. Since V-JEPA and DINOv2 features have different distributions, the DA2 decoder cannot be used directly without adaptation. In our experiments, initializing from the pretrained DA2 head and fine-tuning it worked better than random initialization, so we use this setup in the final model.

Training with the scale-invariant Gaussian NLL loss further improved performance by making the model more robust to heteroscedastic noise, especially in ambiguous or difficult image regions. The uncertainty head therefore adds practical value beyond depth prediction by indicating where the model is less confident.

The main limitation is that the dataset contains only single images, so V-JEPA’s video-native structure is not fully tested. The predictions also suggest that the limited number of valid depth pixels makes direct supervision noisy, motivating a teacher-student setup with denser pseudo-labels. Future work should evaluate the model on video depth data, improve uncertainty calibration, and study distillation for real-time deployment.

5. Summary

We introduced VJEDAI, a hybrid monocular depth model pairing a frozen V-JEPA 2.1 encoder with a Depth Anything-style DPT decoder and a Gaussian uncertainty head, enabling encoder reuse on robotics platforms already running V-JEPA. Our experiments show this is a feasible direction for combining video-native self-supervised representations with dense monocular depth prediction. The per-pixel uncertainty output further adds practical value for generalizability, safe deployment, and future student-model distillation.

References

Chen, S., Guo, H., Zhu, S., Zhang, F., Huang, Z., Feng, J., and Kang, B. Video depth anything: Consistent depth estimation for super-long videos. *arXiv:2501.12375*, 2025.

DepthAnything. Depth-anything-v2. DepthAnything, May 2026. URL <https://github.com/DepthAnything/Depth-Anything-V2>.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., and Houlsby, N. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale, June 2021.

Lin, H., Chen, S., Liew, J., Chen, D. Y., Li, Z., Shi, G., Feng, J., and Kang, B. Depth anything 3: Recovering the visual space from any views, 2025.

Mur-Labadia, L., Muckley, M., Bar, A., Assran, M., Sinha, K., Rabbat, M., LeCun, Y., Ballas, N., and Bardes, A. V-JEPA 2.1: Unlocking dense features in video self-supervised learning, 2026.

Yang, L., Kang, B., Huang, Z., Zhao, Z., Xu, X., Feng, J., and Zhao, H. Depth anything v2. *arXiv:2406.09414*, 2024.